English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics / SEM, 5 September 2024 Examination _ *The grading formula is* 2+ *the number of points. Time allowed: 3 hours. Good luck. We use* m, n *for natural numbers, and* a/b *is equivalent to* $\frac{a}{b}$.

Problem 1. A die has the number 1 on two faces, the number 3 on two other faces, 5 on one remaining face, and 7 on the last face. Let the random variable X be the result of one roll.

- (0.25 points) Find the distribution, expectation and variance of X. Find the probability that in 5 rolls the total number of threes and sevens is 4.
- (0.25 points) Approximately find the probability that the sum of 100 rolls exceeds 2222. What about 4444?
- (0.25 points) Find the expected number of rolls until the hundredth seven. What is the expected number of threes up to that time?

Problem 2. An FMI student, dissatisfied with the unpredictability of standard roulette, decides to create a variant in which past colours affect future ones. Its rules are:

- The roulette wheel has only black and red numbers.
- On spin $n+1$, the probability of a red number is $P_n(m) := \frac{1+m}{2+n}$, where m is the number of red results so far.

Let $X_n := 1_{\{\text{spin } n \text{ produces a red number}\}}$. Given the history $(X_1, \ldots, X_n)$, the conditional distribution of X_{n+1} is Bernoulli with success parameter $P_n(\sum_{i=1}^n X_i)$. For example, if the first results are (red, red, black), then $(X_1, X_2, X_3) = (1, 1, 0)$ and $(P_0, P_1, P_2, P_3) = (1/2, 2/3, 3/4, 3/5)$, so $\mathbb{P}(X_4 = 1 \mid X_1 = 1, X_2 = 1, X_3 = 0) = P_3(2) = 3/5$.

1. (0.25 points) Prove that

$$\begin{aligned}&\mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)\\ &\quad = \mathbb{P}(X_{\sigma(1)} = x_1, X_{\sigma(2)} = x_2, \ldots, X_{\sigma(n)} = x_n)\end{aligned}$$

   for every permutation σ of $\{1, \ldots, n\}$; that is, the vector $(X_1, \ldots, X_n)$ and all its permutations have the same distribution.

2. (0.25 points) Find $\mathbb{P}(X_n = 0)$ and $\mathrm{Cov}(X_i, X_j)$ for $i \neq j$.

3. (0.25 points) Find the distribution of the number of red results W_4 after 4 trials. What is the distribution of W_n after n trials?

4. (0.25 points) Find $\mathbb{P}(X_m = 0 \mid X_n = 1)$ for all m, n.

5. (0.25 points) Find the probability that the nth and $(n+1)$st numbers are equal.

6. (0.25 points) Calculate $\mathbb{P}(W_{n+1} = W_n + 1)$.

7. *Bonus*: (0.5 points) Prove that* P_n *has a limit in distribution, and find it.*

8. *Bonus*: (0.25 points) Propose an algorithm that explicitly constructs a roulette wheel with numbers from 1 to 36, with the even numbers red, and implements the condition above.*

Problem 3. 1. (0.5 points) Let $q \in (0, 1)$ and $U_1 \sim U(0, 1)$. Define

$$X := 1 + \left\lfloor \frac{\ln(U_1)}{\ln(q)} \right\rfloor,$$

   where $\lfloor x \rfloor$ is the integer part of x; for example, $\lfloor 1.15 \rfloor = 1$ and $\lfloor \pi \rfloor = 3$. What values does X take? Find its distribution and expectation.

2. (0.25 points) Let $U_1, U_2 \sim U(0, 1)$ be independent, and let $Y := \ln(U_1)/\ln(U_2)$. Find the density and expectation of Y.

3. (0.25 points) Is it true that $1 + \lfloor \ln(U_1)/\ln(U_2) \rfloor$ has a geometric distribution? Explain why.

Problem 4. 1. (0.25 points) Model the temperature on a given day by $Y = -30 + X$, where $X \sim Exp(\lambda)$. Find λ such that the expected monthly minimum temperature is 10 degrees; assume that a month has 30 days.

2. (0.25 points) Estimate the probability that the average monthly minimum temperature over a period of 10 years exceeds 0.01 degrees. Discuss why this model is or is not good.

3. (0.5 points) Now focus only on summer months. If we model the daily temperature by $Y \sim Exp(\lambda)$, this allows arbitrarily high temperatures, albeit with a small probability. To restrict the values of your model, introduce a random variable Z that has the distribution of Y conditional on Y being between 10 and 40 degrees. Find its density and expectation. What is the expected monthly minimum temperature under the daily temperature model Z?